

Proposal A

Participant u

Proposal B

ID

ID

ID

Learned coordinates (K dimensions)

Separate participant table; one proposal table shared by A and B

Z_A

Z_u

Z_B

Squared Euclidean distance

$$S_A = \|Z_u - Z_A\|^2$$

Squared Euclidean distance

$$S_B = \|Z_u - Z_B\|^2$$

S_A, S_B

Signed difference

$$S_B - S_A$$

Symmetric features

$$(S_A + S_B) / (2K)$$

$$[(S_A - S_B) / K]^2$$

Positive learned multiplier

MLP: $8 \rightarrow 8$ tanh units

$$g = \text{softplus}(\text{output}) + 10^{-6}$$

$\times$

Choice probability

$$P(A) = \text{sigmoid}[(S_B - S_A) g]$$